Preprint. August 12, 2026.

Rediscovering Relativity Is Not a Single Task

What It Takes for AI to Discover a Physical Theory

Dong Zhang

Can an AI system rediscover general relativity? This question is increasingly discussed in the AI community, but it is often framed too narrowly. Some accounts implicitly assume that rediscovery means reproducing Einstein's field equations; others assume that a single batch of experimental data suffices to construct a theory; still others reduce scientific discovery to one mode of inference. We argue that rediscovering a physical theory is not a single task, but a layered sequence of distinct capabilities. Using the history of relativity as a case study, we reconstruct it not as a linear path from Newton to Einstein but as a branching theory space populated by Lorentzian, Poincaréan, scalar, vector, and geometric theories. We show that theory construction and experiment design were mutually generative rather than sequential, and that at several points the available evidence could not adjudicate among live candidates. On this basis we propose a decomposition of rediscovery into distinct capabilities, and three staged tests for AI systems in theoretical physics.

1. Introduction

Can AI rediscover general relativity? This question has recently become a benchmark for evaluating the scientific capabilities of AI systems. One influential proposal is Hassabis's "Einstein test": train a system whose knowledge is restricted to what was known in 1911 and ask whether it can independently arrive at general relativity, just as Einstein did in 1915. Only then, the argument goes, could we claim that the system possesses genuine general intelligence.[1] One such attempt trains models from scratch on corpora containing only pre-1900 texts, aggressively removing any traces of later concepts (Hla, 2026).

The motivation behind these efforts is sound: they try to hide the answer and reduce data contamination. But they still leave the central question underspecified. What counts as "rediscovering" relativity, and how should success or failure be evaluated along the way? This is the gap this paper addresses. In our view, rediscovering relativity is not about recovering a single answer; it is about reconstructing a historical theory space.

So far, the notion of "rediscovery" itself is ambiguous. Does it mean reproducing Einstein's field equation? Does it mean reconstructing the entire logical path from special relativity to general relativity? Or is reproducing only one crucial step sufficient? Different answers correspond to fundamentally different capabilities, yet current discussions often arrive at yes-or-no conclusions without first distinguishing among them.

More importantly, the current discussion often makes the history of relativity look too static, as if all relevant facts had already been collected and the task were to infer the right theory from a fixed archive. The actual history was not shaped in this way. Maxwell's equations predicted electromagnetic waves and incorporated light into electromagnetism, which in turn motivated the search for the so-called "aether wind". The null result of the Michelson–Morley experiment led to Lorentz's contraction hypothesis. The experiments carried out between 1902 and 1904 were no longer testing for the aether wind, but the new predictions generated by the contraction hypothesis itself. The history of relativity was an interaction between theory and experiment: theories suggested what to test next, and experiments forced theories to change. From Maxwell to Lorentz, from Poincaré to Einstein, and later to the competing theories of relativistic gravity, the development of relativity was a continuous process of mutual interaction rather than a one-shot inference.

Instead of discussing rediscovery as one big question, we turn it into three more concrete discovery questions, each closer to the physics and to the historical details:

- Can AI construct and revise theories as new experimental evidence arrives, as Lorentz did?
- Can AI elevate a collection of empirical regularities into a unifying principle, as Poincaré did?
- Can AI construct the entire space of candidate relativistic theories of gravity, derive their predictions, and

dongzhanghz@gmail.com

[1] Demis Hassabis, remarks in "Fireside Chat: Sir Demis Hassabis and Prof Govindan Rangarajan," Indian Institute of Science, 17 February 2026, moderated by Varun Mayya. Transcript available at the IISc EECS event page.

identify the experiments that could distinguish among them?

These are not three versions of the same capability. Together, they ask whether a system can move through a theory space, revise that space as experiments arrive, and identify what must be tested next. The boundary is equally important: theory can tell us what would be informative, but not what observation will show.

Our position is that rediscovering relativity requires reconstructing a historical theory space, repeated interactions between theory and experiment, and movement across different modes of reasoning. It cannot be reduced to induction, deduction, or a single act of abduction, nor can it be completed entirely on paper.

Section 2 reconstructs this theory space. Section 3 reviews three common ways of framing the problem. Section 4 decomposes rediscovery into distinct capabilities. Section 5 proposes three staged evaluations. Section 6 explains why the validation of physical theories must ultimately come from outside the theories themselves.

2. The Historical Theory Space of Relativity

2.1. 19th-Century Situations

Newtonian mechanics and universal gravitation had won enormous authority in 19th-century astronomy. The standard example was Neptune: irregularities in the motion of Uranus were used to infer the position of an unseen planet, which was then observed. Mercury's anomalous perihelion precession was a problem, but not yet a reason to abandon Newtonian gravity. The residual was about $43''$ per century, out of a total precession of roughly $5557''$.

The first responses were patches inside the Newtonian picture. Le Verrier proposed an intra-Mercurial planet, Vulcan, in 1859, and astronomers searched for it for decades without confirmation. Other fixes included solar oblateness and Hall's 1894 modification of the inverse-square exponent. These proposals had weaknesses, but Mercury alone did not force Einstein's answer or open up new physics. As long as old models still had plausible adjustments, a patched Newtonian theory remained the natural first move.

Electrodynamics created a different situation. Maxwell's equations unified electricity, magnetism, and light by treating light as an electromagnetic wave. But the equations were not covariant under Galilean transformations, which made the aether look like the natural rest frame for electromagnetic waves. The Michelson–Morley experiment was designed to detect the aether wind, the motion of the Earth through this frame. Its result was null. By the end of the 19th century, relativity was not born from one clean anomaly. It came out of two live situations: a small gravitational residual inside a very successful Newtonian programme, and an electrodynamic theory whose own structure made the aether experimentally suspect.

2.2. Special Relativity: Lorentz, Poincaré, Einstein

Lorentz's theory lived inside the aether framework. His 1895 theory used local time and length contraction to explain why the Michelson–Morley experiment did not detect aether drift. Later Rayleigh, Brace, and Morley–Miller tested consequences of the FitzGerald–Lorentz contraction idea; Trouton–Noble tested whether a charged capacitor moving through the aether should feel a torque. Lorentz had to update his theory with a more elaborate electron model and additional stabilising assumptions to match these new experiments, but he did not abandon the aether. In this sense, Lorentz was still patching a physical picture.

Compared with Lorentz's increasingly complex theory, Poincaré started from a simpler first principle (he called it "the principle of relativity"). In 1904, he treated the impossibility of detecting aether drift as a general postulate. He corrected Lorentz's formulation, showed that the Lorentz transformations form a group, and made the symmetry requirement explicit. Physically, Poincaré still accepted the aether framework and treated the time and length measured in other frames as artificial quantities. Mathematically, this brought his theory very close to Einstein's.

Unlike the other two, Einstein removed the privilege of the aether and the absolute frame. His 1905 work started from the relativity principle and the constancy of the speed of light, without referring to any hidden rest frame. The time t' of a moving inertial frame was not a calculational fiction, as Poincaré suggested, but the real measurable time of that frame. Today this is the physical interpretation we usually identify with special relativity. But in 1905 there was no experiment that could separate Lorentz, Poincaré, and Einstein by their observable predictions. For the experiments at issue, Lorentz's, Poincaré's, and Einstein's theories gave the same observable results, even though they rested on different physical pictures.

2.3. Theory and Experiment as Mutually Generative

We clarify the interaction between theory and experiment. Between 1887 and 1902, there was only one aether-drift experiment - the Michelson–Morley experiment. Lorentz's 1895 theory of length and local time was good enough to explain the Michelson–Morley result. However, the next experiments were different. Rayleigh and Brace tested whether FitzGerald–Lorentz contraction should produce double refraction in moving matter. Morley and Miller tested whether the effect depended on the material construction of the interferometer. Trouton–Noble tested whether a charged capacitor moving through the aether should feel a torque. These were not just more searches for aether wind. They were new questions created by the theory itself.

All of them gave null results. This is a typical example of theory and experiment moving together: Lorentz's theory explained one experiment, then inspired new experiments, and those experiments pushed him from contraction alone toward the full Lorentz transformation and a more complicated electron theory. After that, no new decisive experiment arrived. Poincaré and Einstein then rebuilt the same experimental situation in cleaner ways, each simpler than Lorentz's physical picture, but neither reformulation was selected by a new observation.

2.4. Relativistic Gravity Before General Relativity

Special relativity is naturally described in Minkowski spacetime, but Newtonian gravity does not fit there. Poisson's equation is not Lorentz covariant. How should gravity be brought into Minkowski spacetime? Between 1905 and 1915 this was an open problem worked on by Poincaré, Nordström, Abraham and Mie, not by Einstein alone. The standard move was to keep spacetime flat and add a gravitational field on it, exactly as one adds an electromagnetic field. The field has to be Lorentz covariant, it has to reduce to Newtonian gravity in the static weak-field limit, and its source has to be the energy-momentum tensor $T_{\mu\nu}$.

Poincaré took the vector in 1905, modelling gravity on electromagnetism, where the source is a four-current j^μ and the field a four-potential A^μ. It was the only template available, since the energy-momentum tensor did not yet exist as a tool. It gave gravitation a propagation speed c and what Poincaré called gravitational waves, but little else to test with observation: a perihelion advance for Mercury well short of the observed $43''$, and no other prediction.

Nordström kept Newton's potential as a scalar and kept it as a field on Minkowski spacetime. Newton's equation contains derivatives in space only, which is why its gravity acts everywhere at once. Nordström added the missing time derivative, which turns it into a wave equation and makes the potential travel at c, and he replaced mass density as the source by the energy of matter, since in relativity the two are the same thing. His theory was self-consistent and attractive, and it explicitly requires $m_i = m_g$, so the equivalence principle was not Einstein's private concern, and the gravitational redshift comes out right for that reason. However, he calculated a perihelion precession of $-7''$ per century for Mercury, with the sign reversed, and no bending of light passing through the gravitational field of matter.

Einstein started on the same flat background, and with the same choice as Nordström. From 1907 to 1912 he worked with a scalar, using a position-dependent speed of light $c(x)$, and the $0.83''$ deflection he published in 1911 came from it, half of the value now measured. He gave the scalar up when it could not be carried beyond static fields or made to hold the full equivalence principle.

What he turned to was not another field placed on Minkowski spacetime. He took gravity to be the geometry of spacetime itself, and that settles the rank without any further choice: geometry is carried by the metric in $ds^2 = g_{\mu\nu} dx^\mu dx^\nu$, and the metric is a symmetric second-rank tensor by construction. The rank was not selected. It was handed over by the geometry.

Having the idea was not having the theory. The field equations took three more years and arrived in November 1915, and only then did the predictions come out right: a perihelion precession of $+43''$ per century for Mercury, and a deflection of $1.75''$ for light grazing the Sun, twice the value he had published in 1911.

3. Common Framings of Rediscovery

3.1. The Whig History Framing

Whig history writes the past from the "winning end". It keeps the line that leads to the present, drops the branches that left no descendants, and in doing so makes the outcome look like the only thing that could have happened. Applied to relativity, it records Einstein and his reasoning alone, and omits the parallel work of Lorentz, Poincaré, Nordström, Abraham, Mie and so on.

The resulting account is familiar. Special relativity rests on two postulates, the principle of relativity and the constancy of the speed of light. General relativity begins from an anomaly, the $43''$ per century left unexplained in Mercury's perihelion, proceeds through the equivalence principle, takes Riemannian geometry as its mathematical tool, and arrives at Einstein's field equations coupling matter to the geometry of spacetime, which displaces the Newtonian paradigm.

As a framing of the discovery problem, it carries a further assumption: at each historical moment, there was already one definite correct answer, and discovery meant reaching it. This assumption makes the framing operational. An AI evaluation built on it can score a system by asking whether it outputs the two postulates of special relativity, or the Einstein field equations, and count anything else as failure. This framing appears in the Einstein test quoted in Section 1, in benchmark designs that train on pre-1900 corpora and then check for traces of later concepts (Hla, 2026), and in popular accounts that follow Einstein alone.

3.2. The One-Shot Framing

On this view, once the data are in place, the remaining task is to apply the right form of inference and get the theory out. The pipeline is linear: experiments first produce data, the data are then processed, and a theory comes out at the end. Discovery is the processing step. The theory works on an archive that is already in place when the work begins.

Applied to relativity, this account says the following. By the turn of the century, the experiments on the propagation of light were already in hand, and special relativity follows from them. Mercury's anomalous perihelion precession was also already known from nineteenth-century astronomy. Once special relativity is available, that old discrepancy becomes the remaining input from which general relativity follows. Each theory appears in a single step, as the correct reading of the evidence available at the time.

Moved into AI, the framing gives a clean experimental design: give a system all of physics up to 1905 and have it produce general relativity. It appears in public statements by the leadership of AI laboratories, and in benchmark designs that supply a complete corpus in a single prompt and then check whether the target theory is present in the output. It presupposes that the evidence base is fixed before theorising begins, so that all of discovery can be placed in one inference from data to theory.

3.3. The Single-Mechanism Framing

On this view, scientific discovery comes down to a single mode of inference. There is one operation that turns what is already known into a theory. Building a discovery system then means finding that operation and implementing it well. Different accounts name different operations, and each treats the operation it names as the whole task.

Applied to relativity, the episode becomes a case where a single mode of inference worked. In the inductive reading, Einstein found the shortest description covering the observations, and the theory is the compressed form of the data. In the deductive reading, the postulates were already in place and the theory is what follows from them, so the work is derivation. In the abductive reading, the theory arrived in a single leap to the best available explanation, and what came afterwards was elaboration.

Each reading has a current research programme attached to it. The compression line runs through Schmidhuber's treatment of discovery as compression progress. The deductive line runs through formal mathematics and systems such as AlphaProof, where results are obtained by searching a fixed set of axioms. The abductive line treats discovery as one explanatory jump and holds in addition that such a jump requires embodied simulation. What the framing presupposes is that a single mechanism accounts for the whole episode, so that progress on that mechanism is progress on discovery itself.

4. Rediscovery Is Not One Task

Two facts often placed at the origin of relativity did not by themselves produce it. The equality of inertial and gravitational mass had been known for two centuries, but most theories simply had to respect it. It was not yet treated as the problem to solve. Mercury's unexplained $43''$ per century was a real discrepancy, but Newtonian fixes could still absorb it: an undiscovered planet, a flattened Sun, or a small change to the exponent in the inverse-square law. The new framework came from a different pressure: Newtonian mechanics and Maxwell's electrodynamics did not fit together. Maxwell's equations single out a speed and are not covariant under Galilean transformations, while experiments failed to detect the aether wind that the old picture expected.

That incompatibility opened the problem rather than settling it. What followed took sixty years, and it was not a single step: an explanation of the null result, its revision under a second round of experiments, a principle, special relativity, then the separate problem of making gravity Lorentz covariant, a space of candidate theories, and finally an observation that chose among them. Table 1 sets out the chain.

Two features of the table matter more than any single row. The first is that theory and experiment alternate. Tasks 1 and 3 are experiments, and task 3 exists only because task 2 was carried out: Rayleigh, Brace, Morley–Miller and Trouton–Noble were testing consequences that Lorentz's contraction hypothesis had itself generated. The evidence base was not fixed in advance and then read; part of it was manufactured by the theories under test.

The second is that no task's output was determined by the task before it. The same body of evidence supported Lorentz's, Poincaré's and Einstein's formulations at task 6, and supported scalar, vector and tensor theories of gravity at task 8a. Tasks that look like a single act of inference from the outside were, in each case, a choice among live alternatives that the evidence did not settle.

The rows divide on one criterion: whether the work is done on paper or in the world. Tasks 2 and 4 through 8b are paper work, and there is no reason in principle why a system could not perform them. Tasks 1, 3, 9, 10 and 11 are not. A system can specify what should be measured, with what apparatus and to what precision, and identifying that light deflection rather than redshift is the quantity that separates the candidates is itself a theoretical result. What it cannot do is supply the answer. Redshift is the sharpest illustration: Nordström's scalar theory gives the correct value, so confirming it in 1959 discriminated nothing, and the entire adjudication at task 9 rested on a number that no amount of reasoning could produce in advance.

Table 1. The relativity episode decomposed into tasks. Shaded rows are the tasks that have to be settled in the world. "Design only" means an AI can specify the experiment, down to apparatus and required precision, but a human has to build it and run it, and the answer comes from the world rather than from the reasoning.

#	Task	Capability it requires	Can AI attempt it?
0	Newtonian mechanics and Maxwell's electrodynamics	starting point	—
1	Michelson–Morley: measure the speed of the aether wind	turning a theoretical question into an apparatus	design only
2	Lorentz 1.0: contraction and local time	theory construction on incomplete evidence	yes
3	Rayleigh, Brace, Morley–Miller, Trouton–Noble	deriving new testable questions from a theory	design only
4	Lorentz 2.0: the full transformation	revision under a second round of null results	yes
5	Poincaré: the relativity principle	principle formation from accumulated patches	yes
6	Einstein: special relativity	reformulation without a privileged frame	yes
7	Pose the next problem: make gravity Lorentz covariant	problem recognition	yes
8a	Scalar, vector and tensor candidates	hypothesis-space construction	yes
8b	Predictions for each candidate	derivation and screening on paper	yes
9	Eddington 1919: deflection of starlight	identifying the discriminating observation	design only
10	Gravitational redshift confirmed, 1959	later confirmation, does not discriminate	design only
11	Gravitational waves detected, 2015	later confirmation	design only

5. Three Staged Tests

- These are conceptual tests, not benchmarks ready to run. The answers are present in any modern training corpus, so success is uninformative; only failure is informative, and only counterfactual variants, altered source symmetry or altered spacetime dimension, restore any diagnostic value.

5.1. The Lorentz Test

- Input, staged: first Maxwell plus aether plus Michelson–Morley alone; then Rayleigh/Brace, Trouton–Noble, Morley–Miller.

- Tasks: construct an explanation; derive its new predictions; on receiving the second batch, decide whether to patch, abandon, or reformulate.

- What would count as success: deriving from one's own hypothesis the predictions that the second batch will test; recognising that those predictions have been refuted; tracking the growth in auxiliary assumptions.

5.2. The Poincaré Test

- Input: the full set of null results and Maxwell's equations. Withheld: the relativity principle, the Lorentz group, Einstein's postulates.

- Tasks: find a principle that explains all results at once; derive the admissible transformations; establish closure.

- What would count as success: the group property, and an explicit statement of why a principle differs from a sequence of compensations.

5.3. The Gravity Test

- Input: Minkowski spacetime; Lorentz covariance; $\partial^{\mu} T_{\mu\nu} = 0$; recovery of Newtonian gravity in the static weak field. Withheld: equivalence principle, Riemannian geometry, spin-2 gravity, Einstein, general relativity.

- Tasks: enumerate candidate field types with justification; write couplings and field equations; compute Newtonian limit, perihelion precession, light deflection, redshift; separate paper-eliminable from observation-requiring candidates; name the decisive observation.

- What would count as success: the enumeration is complete, spin 0, 1, 2; vector is eliminated without computation; the decisive observable is identified as light deflection with the reason why redshift and perihelion do not serve.

- Table of the three candidates and their predictions.

6. Why Physics Must Remain Coupled to the World

- Formal proof certifies mathematical validity internally; AI for mathematics can in principle close its loop.

Table 2. Candidate relativistic gravity theories and discriminating predictions.

Candidate Field Type	Mercury Precession	Light Deflection	Gravitational Redshift	Outcome
Spin 0, scalar (Nordström)	$-7''$ per century, sign reversed	0, spacetime conformally flat	correct	survives on paper, eliminated by observation
Spin 1, vector (Poincaré)	advance well short of $43''$	none predicted	none predicted	eliminated without computation, like sources repel
Spin 2, tensor (Einstein)	$+43''$ per century	$1.75''$	correct	survives

- Physics cannot. Simulation does not supply independent evidence: it executes the theory under test. Computing $1.75''$ from general relativity and then citing that number as confirmation of general relativity is circular. The 1919 measurement supplied an input from outside the theory.

- The loop: theory $\rightarrow$ prediction $\rightarrow$ experiment $\rightarrow$ observation $\rightarrow$ revision.

- Execution may be automated, including self-driving laboratories and robotic observatories. The claim is not about who operates the instrument. The claim is that the validating information must originate outside the theory being tested.

7. Discussion and Conclusion

- Restate: the question is not whether a system reaches Einstein's answer.

- The decisive test is whether it can reconstruct the space of possibilities, recognise what remains undecidable on paper, and determine which question must next be put to nature.

- Where the division of labour falls: enumeration and paper-elimination on one side, adjudication on the other.

- Limitations: a single case; the enumeration in Section 5.3 uses representation-theoretic tools unavailable before 1939; the tests are conceptual.

References

Hla, M. Machina mirabilis. `https://michaelhla.com/blog/machina-mirabilis.html`, March 2026. Online research note.

A. Appendix A

- A.1 The three candidate gravity theories: couplings, field equations, and derivations of the four observables.

- A.2 Timeline of experiments and theories, 1859–1919.